2.1 The physical processes that shape the coast

This chapter will explain the causes and consequences of coastal erosion:

★ transportation
★ deposition
★ longshore drift
★ types of wave
★ wave refraction.

▲ **Figure 2.1** Jolly Harbour, Antigua

The factors that affect coastal processes and coastal landforms include:

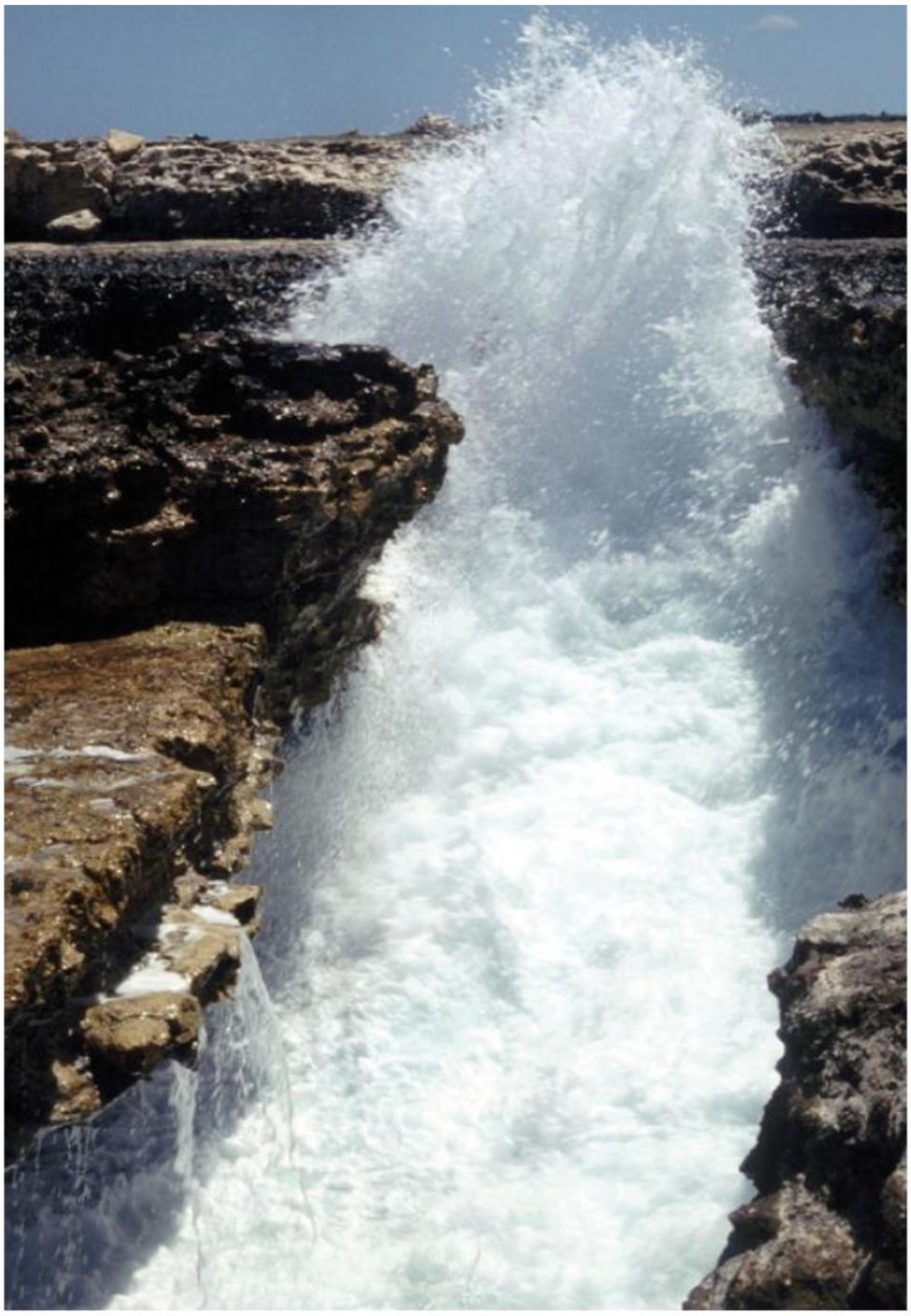
» waves and currents, including longshore drift
» local geology – that is, rock type, structure and strength
» changes in **sea level**
» human activity and the increased use of coastal engineering.

All of these factors interact and produce a unique set of processes that occur at the coast. These processes go on to produce different types of landform for every coastal area.

Coastal erosion

There are many types of erosion carried out by **waves**:

» Hydraulic action occurs as waves hit or break against a cliff face. Any air trapped in cracks is put under great pressure. As the wave retreats, this build-up of pressure is released with explosive force (Figure 2.2). This is especially important in well-jointed rocks such as limestone, sandstone and granite, and in weak rocks such as clays and glacial deposits. Hydraulic action makes the most impact during storms.
» Corrasion (abrasion) is the process of a breaking wave hurling materials, such as pebbles or shingle, against a cliff face. It is similar to abrasion in a river.



▲ **Figure 2.2** Hydraulic action on a coral coastline

- » Attrition is the process by which eroded material, such as broken rock, is worn down to form smaller, rounder beach material.
- » Corrosion (solution) occurs on limestone and chalk. Calcium carbonate, a salt found in these rocks, dissolves slowly in acidic water.

Transportation

Transportation in coastal systems is generally divided into two types:

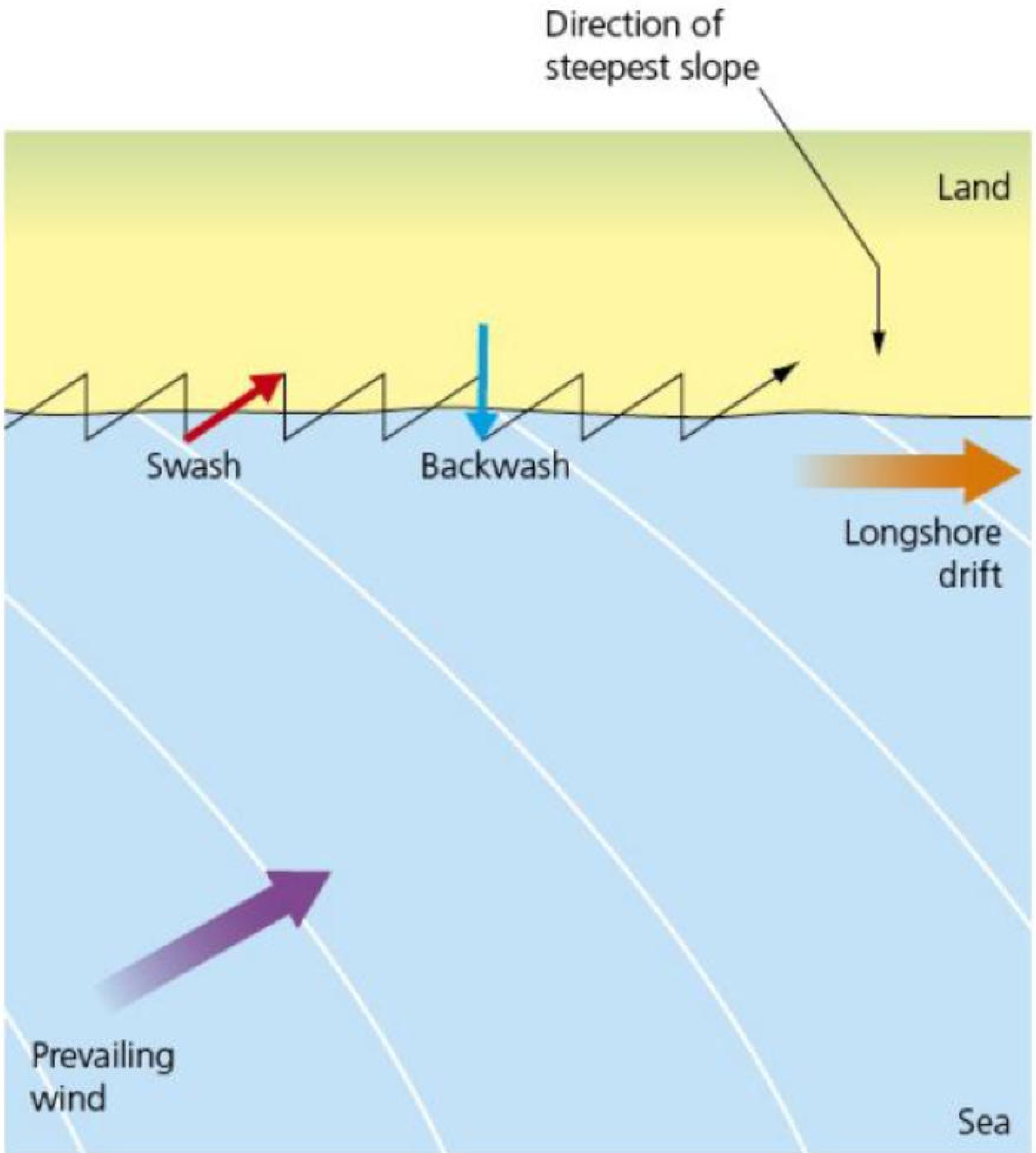
- » Bedload: Grains transported by bedload are moved through continuous contact (traction), dragging or discontinuous contact (saltation or bouncing) with the sea floor. In traction, sediments slide or roll along the sea floor – a slow form of transport. Weak currents may transport sand, and strong currents may transport pebbles and boulders. In saltation, the grains of sediment bounce along the sea bed. Moderate currents may transport sand, whereas strong currents may transport pebbles and gravel.
- » Suspended load: Grains are carried by turbulent flow and generally held up by the water. Suspension occurs when moderate currents are transporting sands. Some grains are permanently held in suspension (wash loads), and these typically consist of clay.

Deposition

Deposition is governed by sediment size and shape. In some cases, notably clay, sediments will stick together (flocculate), become heavier and be deposited. Deposition occurs when there is a decrease in wave energy or velocity. This may occur along a gently sloping shoreline where there is friction with the sea bed, or when waves meet an irregular, indented coastline, for example at the mouth of a river. Obstacles on a beach, such as **groynes**, will cause a wave to break and may lead to deposition updrift of the groyne.

Longshore drift

Longshore drift occurs when waves move up to the beach (**swash**) in one direction, but the waves draining back down the beach (**backwash**) take a different route (under the effect of gravity). The net movement is therefore *along* the shore, hence the term 'longshore drift' (Figure 2.3 and Figure 2.4). A wooden or concrete wall (groyne) may be built to prevent longshore drift moving sand or shingle away from the beach.



▲ **Figure 2.3** Longshore drift

▲ **Figure 2.4** Longshore drift has caused the build up of sediment on the downdrift (near side) of the stone groyne and its removal on the updrift (far side) of the groyne

Human activity and longshore drift in West Africa

The increase in coastal retreat in Ghana has been blamed on the construction of the Akosombo Dam on the Volta River. It is just 110 km from the coast and disrupts the flow of sediment from the Volta, stopping it from reaching the shore. Thus there is less sand to replace that which has already been washed away by longshore drift, so the coastline is retreating due to erosion by the Guinea Current. Towns such as Keta, 30 km east of the Volta estuary, have been destroyed as their protective beach has been removed (Figure 2.5).

▲ **Figure 2.5** Human activity and longshore drift in West Africa

Types of wave

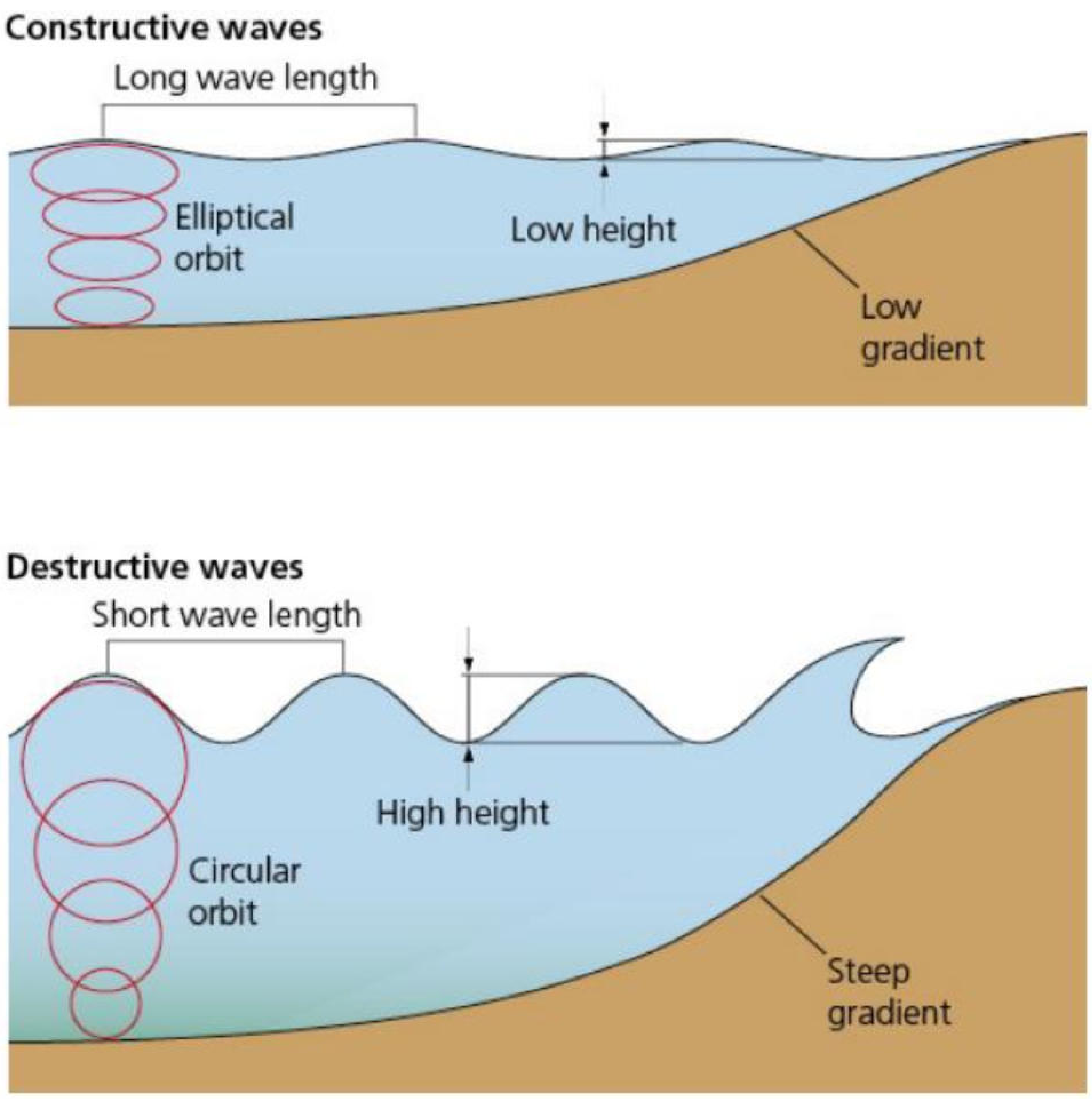
Wave length is the distance between two successive crests or troughs. Wave height is the distance between the trough and the crest. Wave frequency is the number of waves per minute. Wave velocity is the speed of a travelling wave, and is influenced by wind, **fetch** and depth of water. The fetch is the amount of open water over which a wave has passed. Waves are sometimes divided into **constructive** and **destructive waves** (Table 2.1 and Figure 2.6).

▼ **Table 2.1** Destructive and constructive waves

Destructive waves (erosional waves)	**Constructive waves (depositional waves)**
Short wave length (less than 20 m)	Long wave length (up to 100 m)
High height (more than 1 m)	Low height (less than 1 m)
High frequency (10–12/minute)	Low frequency (6–8/minute)
Low period (one every 5–6 seconds)	High period (one every 8–10 seconds)
Backwash greater than swash	Swash greater than backwash
Steep gradient	Low gradient
Caused by local winds and storms	Caused by swell from distant storms
High-energy waves	Low-energy waves

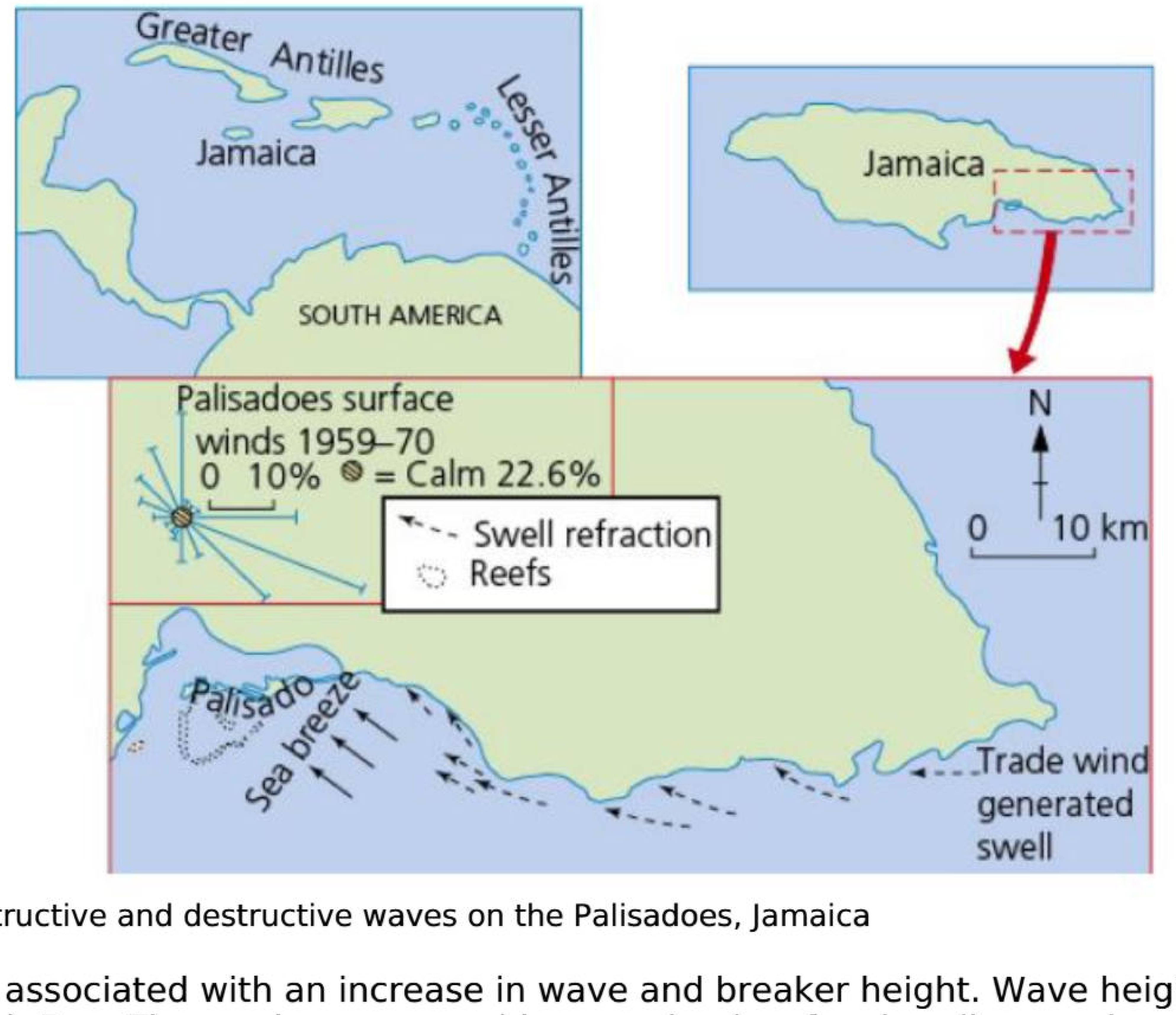
Waves on the Palisadoes, Jamaica

Beaches are transitory features – they regularly change shape. In Jamaica, one of the most important factors for wave formation is the local sea–land breeze. The Palisadoes, on the south coast of Jamaica, has a tidal range of just 0.23 m.



▲ **Figure 2.6** Destructive and constructive waves

Here, swell is generated by trade winds on a year-round basis, and occasionally by easterly waves and tropical cyclones (hurricanes) that originate far from the island. The sea breeze is most persistent in the summer months from May to August, and strongest in June. It most commonly approaches from an east-southeasterly direction (Figure 2.7). The breeze normally develops in the late morning, reaching velocities of 12 m/s by the early afternoon. Once the sea breeze declines, a land breeze develops from the northwest.



▲ **Figure 2.7** Constructive and destructive waves on the Palisadoes, Jamaica

The sea breeze is associated with an increase in wave and breaker height. Wave heights regularly exceed 1 m and may reach 5 m. The sea breezes provide a mechanism for shoreline erosion, caused by destructive waves during daylight hours. The change from constructive waves, where sediment is transported landwards, to destructive waves, where sediment is dragged seawards, is related to wave steepness. As steepness increases, erosion occurs.

When onshore winds occur, a return counter-current is formed, flowing seawards towards the break point. This current removes material from the front of the beach and carries it to the break point, where material is deposited to form longshore bars offshore from the Palisadoes.

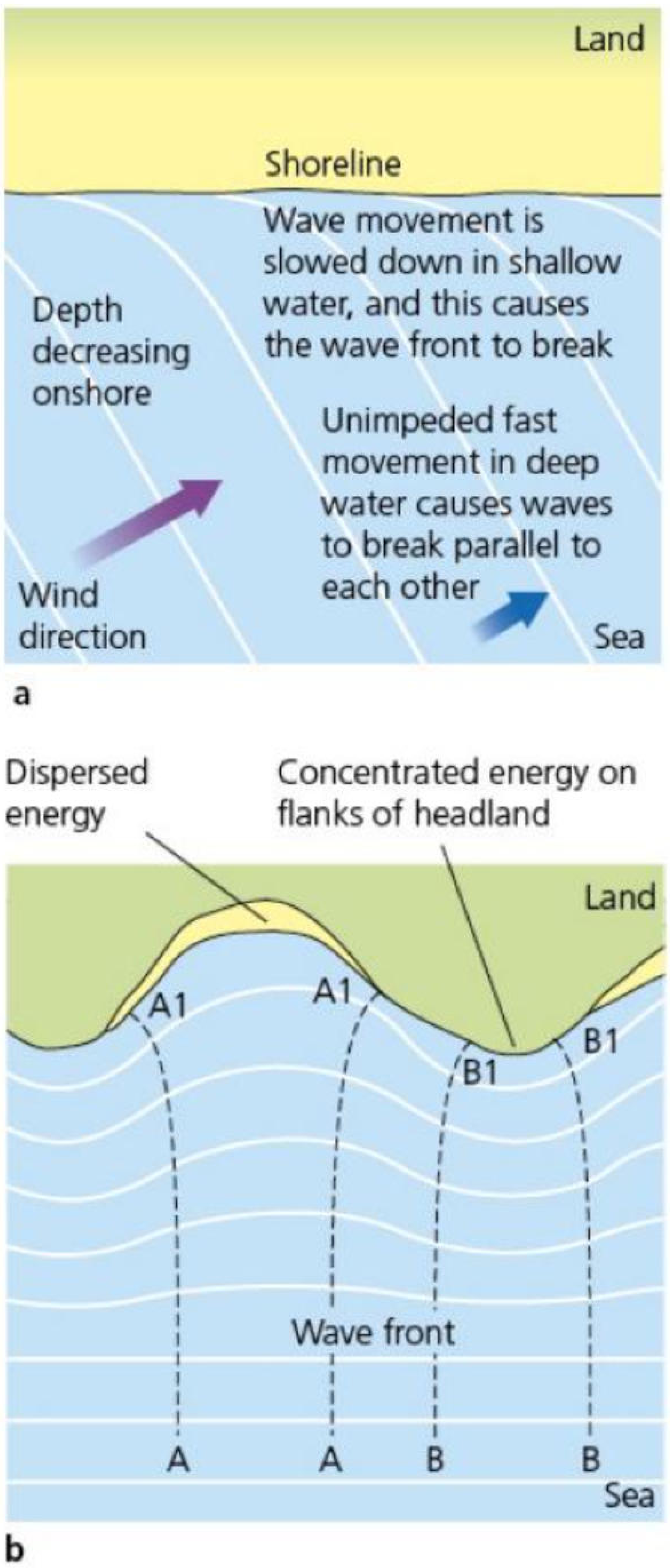
With the return of the land breeze, the destructive waves decay. The swell generated by the trade wind becomes the dominant wave, returning sediment lost from the beach during the day.

Activities

1 State the size of the tidal range of the Palisadoes.
2 Identify the direction in which the sea breeze blows.
3 Identify the direction in which the land breeze blows.
4 Suggest the impacts of:
  a the sea breeze
  b the land breeze
  on wave activity.

Wave refraction

Waves result from friction between wind and the sea surface. Waves in the open, deep sea are different from those breaking on shore. Sea waves are forward-moving surges of energy. Although the shape of the surface wave appears to move, the water particles follow a roughly circular path within the wave. As waves approach the shore, their speed is reduced as they touch the sea floor (Figure 2.8). **Wave refraction** causes two main changes: the speed of the wave is reduced, and the shape of the wave front is altered. If refraction is completed, the wave fronts will break parallel to the shore.



▲ **Figure 2.8** Wave refraction

Wave refraction also distributes wave energy along a stretch of coast. On a coastline with alternating **headlands** and bays, wave refraction will concentrate destructive/erosive activity on the headlands, while deposition will tend to occur in the bays. Irregularities in the shape of the coastline mean that refraction is not always totally achieved. This causes longshore drift**,** which is a major force in the transport of material along the coast (Figure 2.3 on page 36).

Activities

1 Define the following terms:
- Swash
- Fetch
- Wave refraction
- Backwash.

2 Describe the main differences between a destructive wave and a constructive wave.
3 Describe and explain the process of longshore drift.
4 Briefly describe how human activity affected the impact of longshore drift in West Africa.